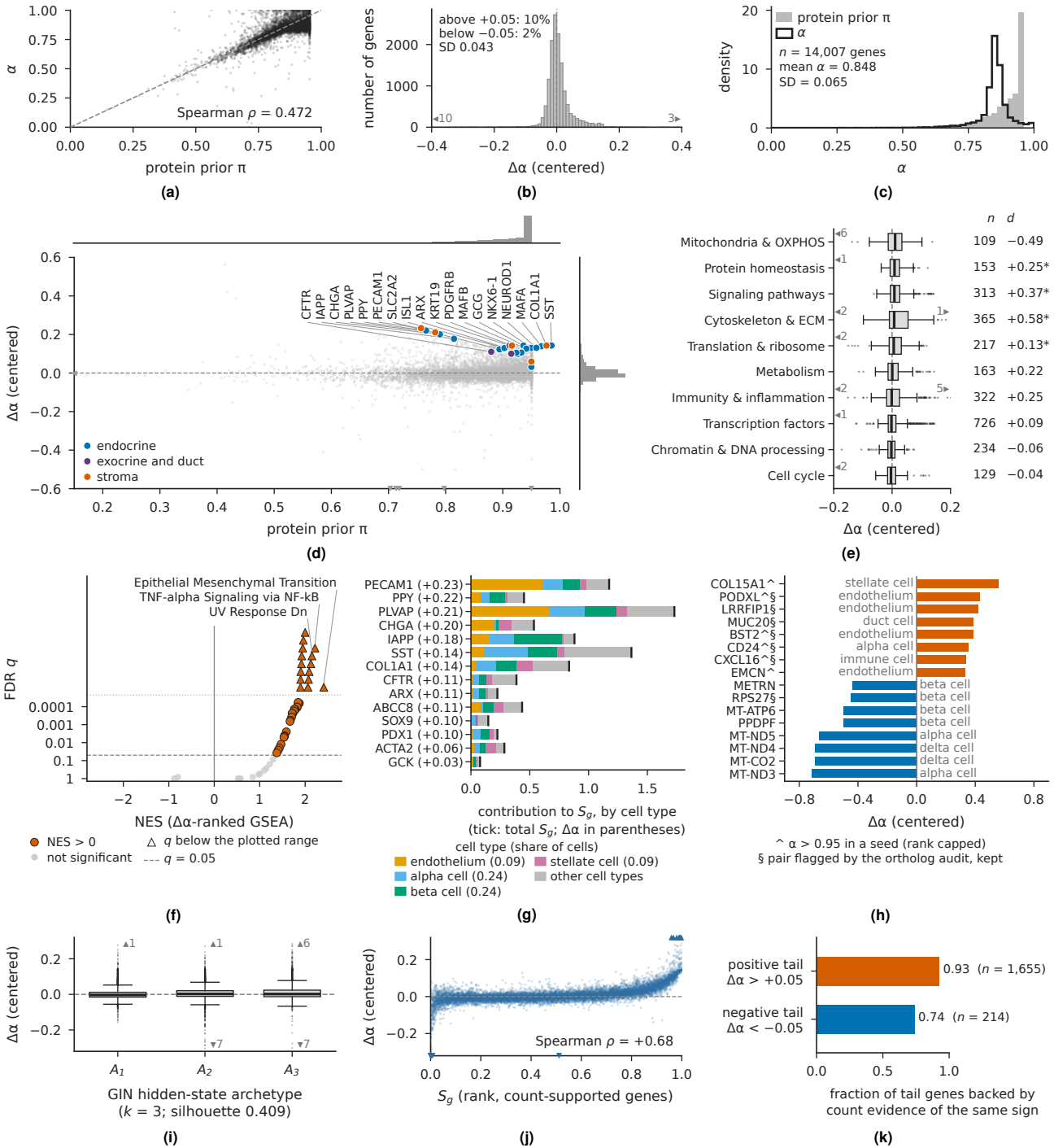

Spearman ρ = 0.472
protein prior π
α
(a)
above +0.05: 10%
below −0.05: 2%
SD 0.043
number of genes
Δα (centered)
(b)
protein prior π
α
n = 14,007 genes
mean α = 0.848
SD = 0.065
density
(c)
CFTR
IAPP
CHGA
PLVAP
PPY
PECAM1
SLC2A2
ISL1
ARX
KRT19
PDGFRB
MAFB
GCG
NKX6-1
NEUROD1
MAFA
COL1A1
SST
endocrine
exocrine and duct
stroma
Δα (centered)
protein prior π
(d)
n
d
Mitochondria & OXPHOS 109 −0.49
Protein homeostasis 153 +0.25*
Signaling pathways 313 +0.37*
Cytoskeleton & ECM 365 +0.58*
Translation & ribosome 217 +0.13*
Metabolism 163 +0.22
Immunity & inflammation 322 +0.25
Transcription factors 726 +0.09
Chromatin & DNA processing 234 −0.06
Cell cycle 129 −0.04
Δα (centered)
(e)
Epithelial Mesenchymal Transition
TNF-alpha Signaling via NF-kB
UV Response Dn
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q below the plotted range
q = 0.05
(f)
PECAM1 (+0.23)
PPY (+0.22)
PLVAP (+0.21)
CHGA (+0.20)
IAPP (+0.18)
SST (+0.14)
COL1A1 (+0.14)
CFTR (+0.11)
ARX (+0.11)
ABCC8 (+0.11)
SOX9 (+0.10)
PDX1 (+0.10)
ACTA2 (+0.06)
GCK (+0.03)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
endothelium (0.09)
stellate cell (0.09)
alpha cell (0.24)
other cell types
beta cell (0.24)
(g)
COL15A1^ stellate cell
PODXL^§ endothelium
LRRFIP1§ endothelium
MUC20§ duct cell
BST2^§ endothelium
CD24^§ alpha cell
CXCL16^§ immune cell
EMCN^ endothelium
METRN beta cell
RPS27§ beta cell
MT-ATP6 beta cell
PPDPF beta cell
MT-ND5 alpha cell
MT-ND4 delta cell
MT-CO2 delta cell
MT-ND3 alpha cell
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
A_1
A_2
A_3
GIN hidden-state archetype
(k = 3; silhouette 0.409)
(i)
Spearman ρ = +0.68
S_g (rank, count-supported genes)
Δα (centered)
(j)
positive tail
Δα > +0.05
0.93 (n = 1,655)
negative tail
Δα < −0.05
0.74 (n = 214)
fraction of tail genes backed by
count evidence of the same sign
(k)